Kernix 0.13.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

51 workloads

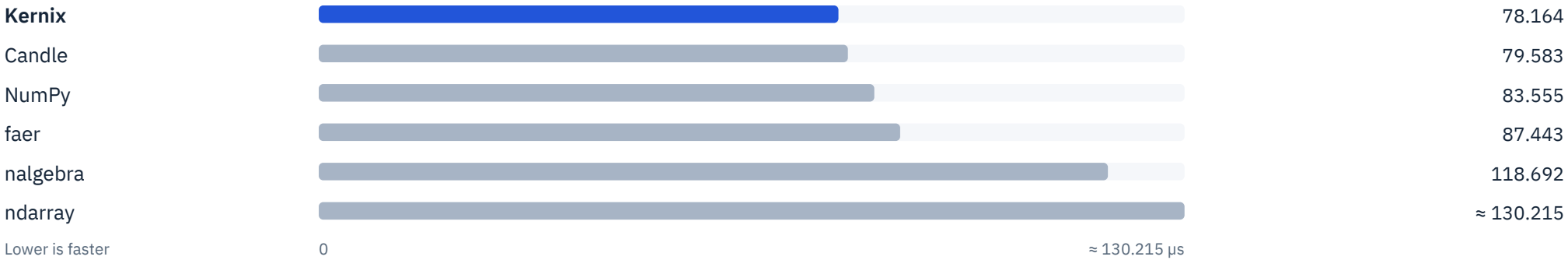
175 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.13.0 / Sparse matrices & memory

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse matrix × vector

Per-call median

2048 × 2048 · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Kernix 0.13.0 / Sparse matrices & memory

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f32 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × sparse matrix

Per-call median

256 × 256 · five diagonals each · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



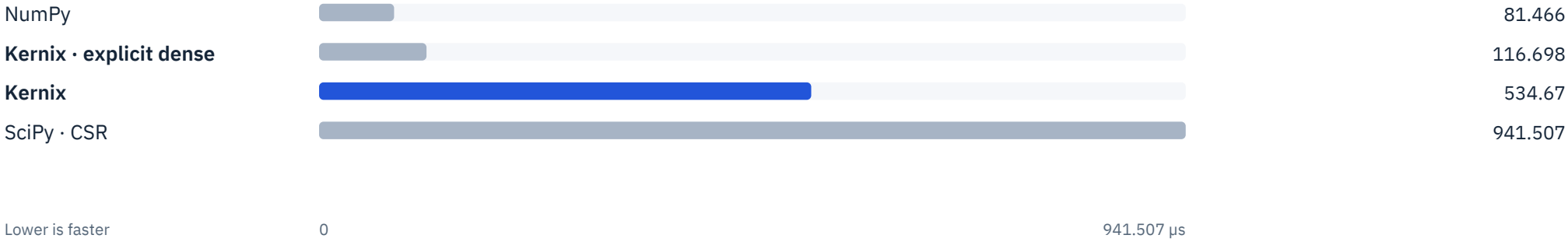
Kernix 0.13.0 / Sparse matrices & memory

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Automatic CSR construction

Per-call median

512 × 512 · clone + scan + compress · f64 · Construction · dense copy or compression · preparation costs included



Kernix 0.13.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



Kernix 0.13.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



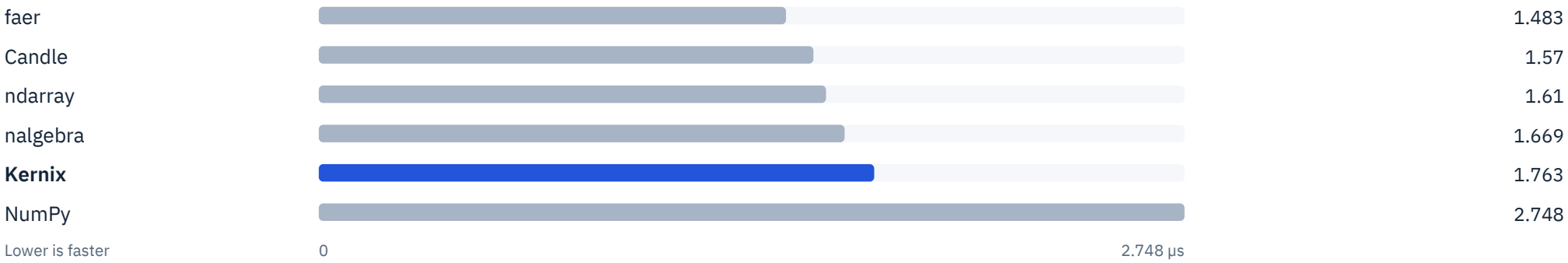
Kernix 0.13.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

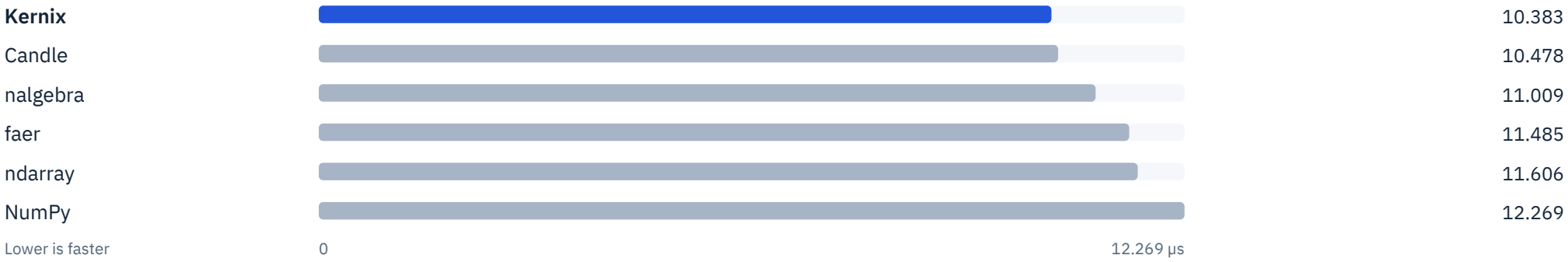
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



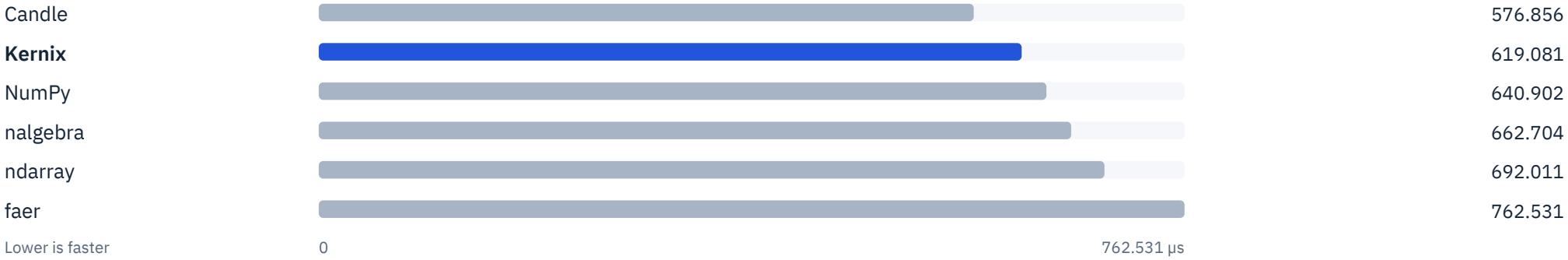
Kernix 0.13.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

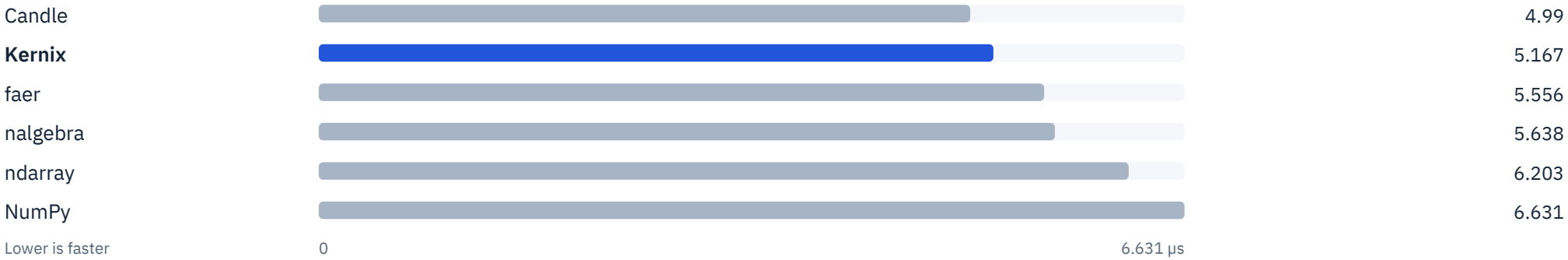
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



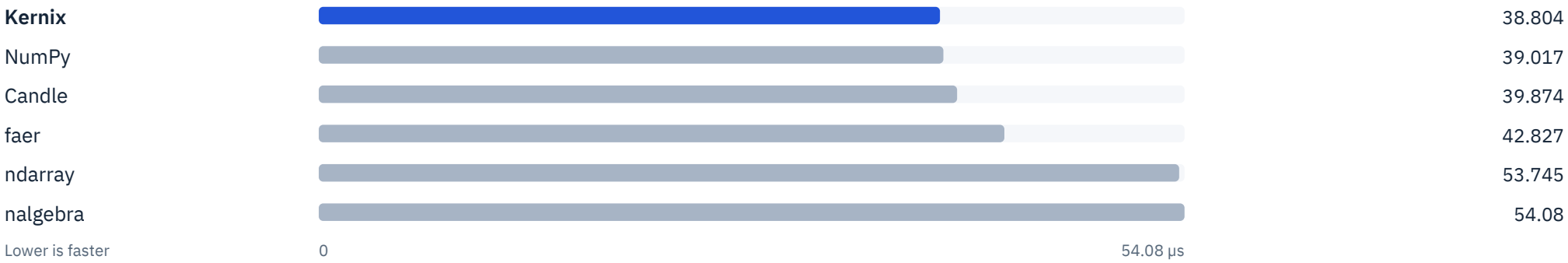
Kernix 0.13.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

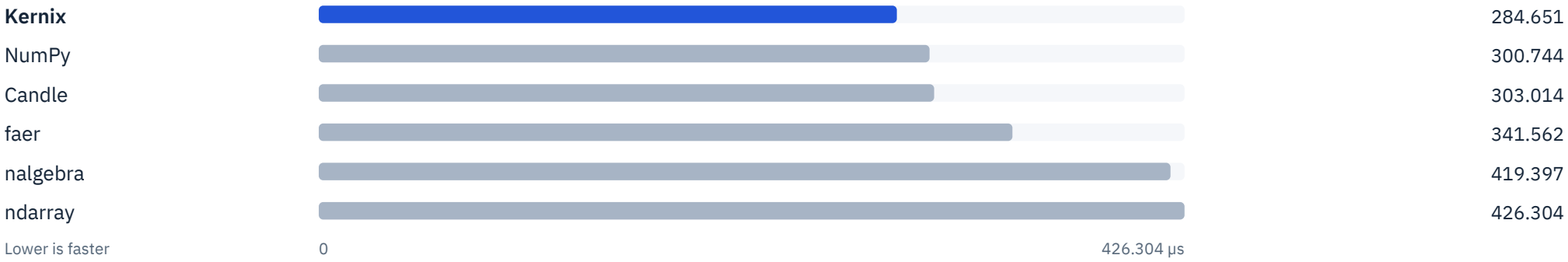
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



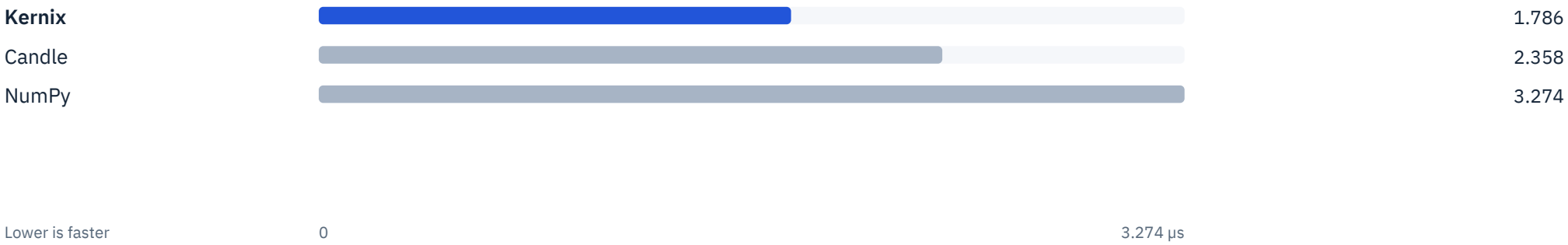
Kernix 0.13.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

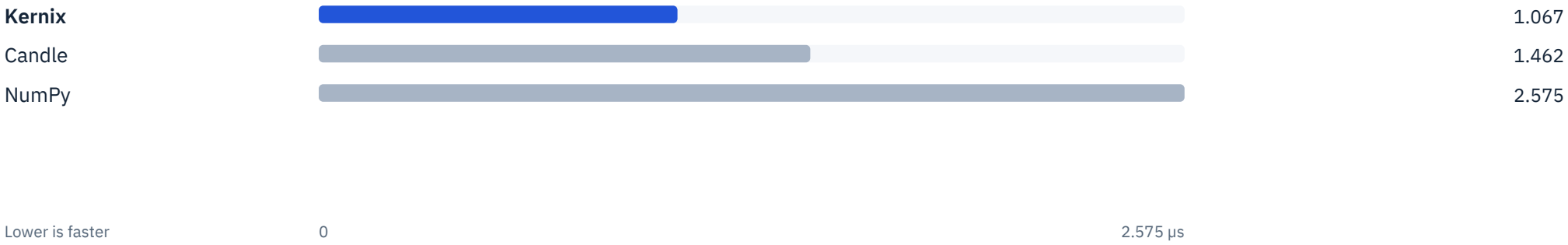
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.13.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.13.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

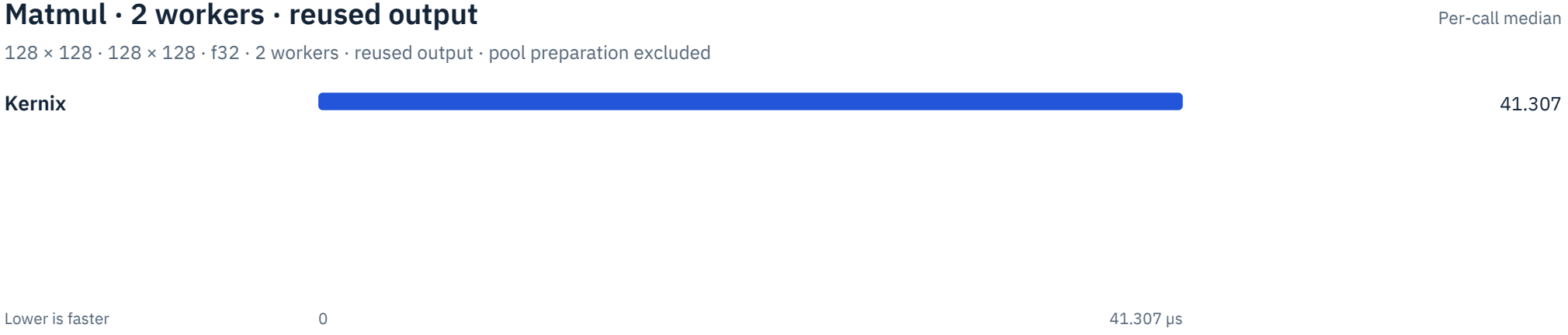
Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



Kernix 0.13.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Kernix 0.13.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



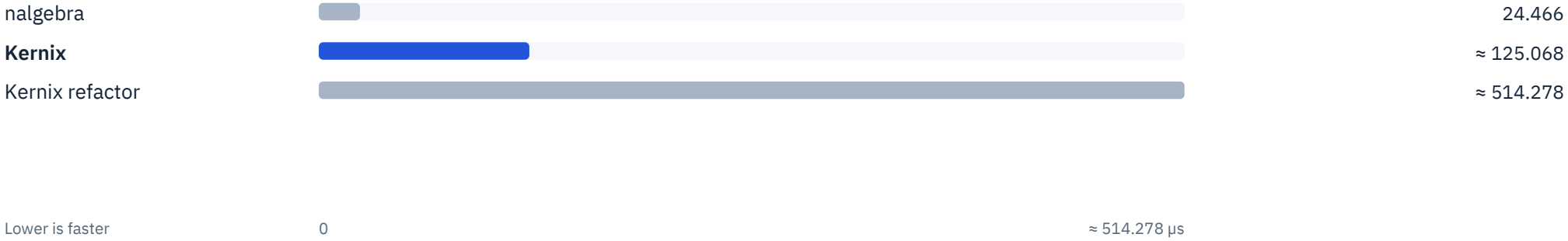
Kernix 0.13.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

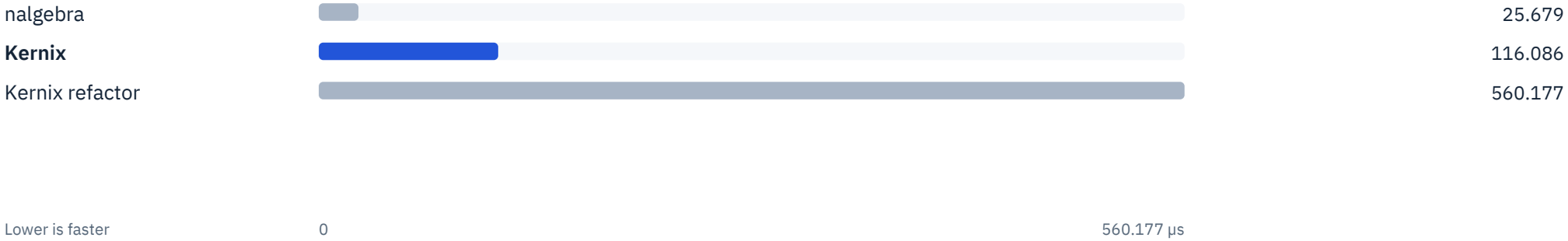
256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

Per-call median

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



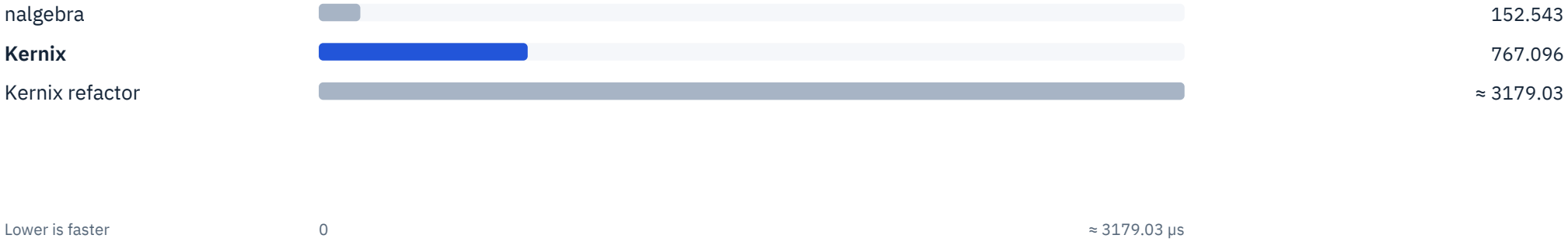
Kernix 0.13.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Thin QR

Per-call median

512 × 128 · f64 · 1 worker · allocated output



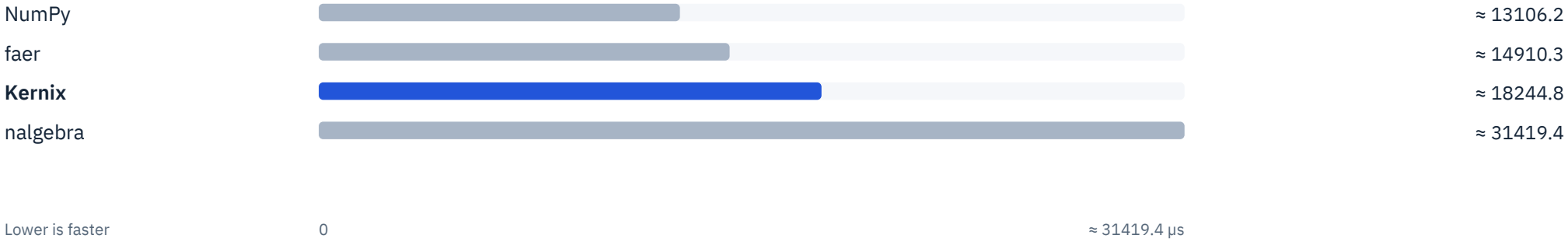
Kernix 0.13.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

Per-call median

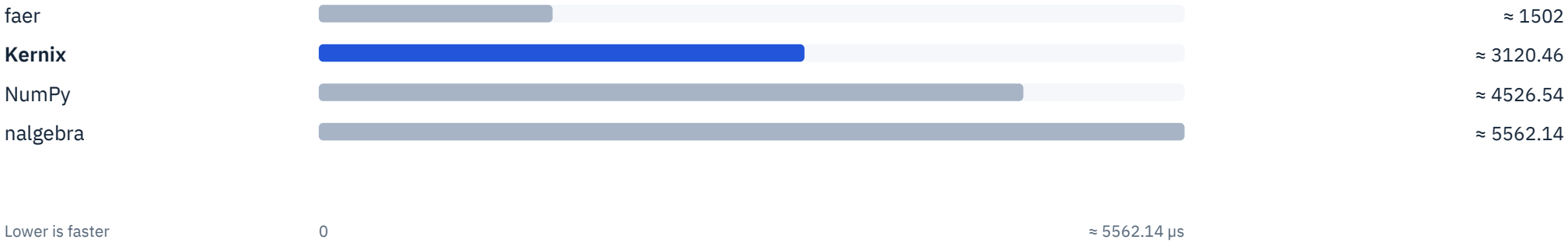
1024 × 256 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

512 × 512 · f64 · 1 worker · allocated output



Kernix 0.13.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

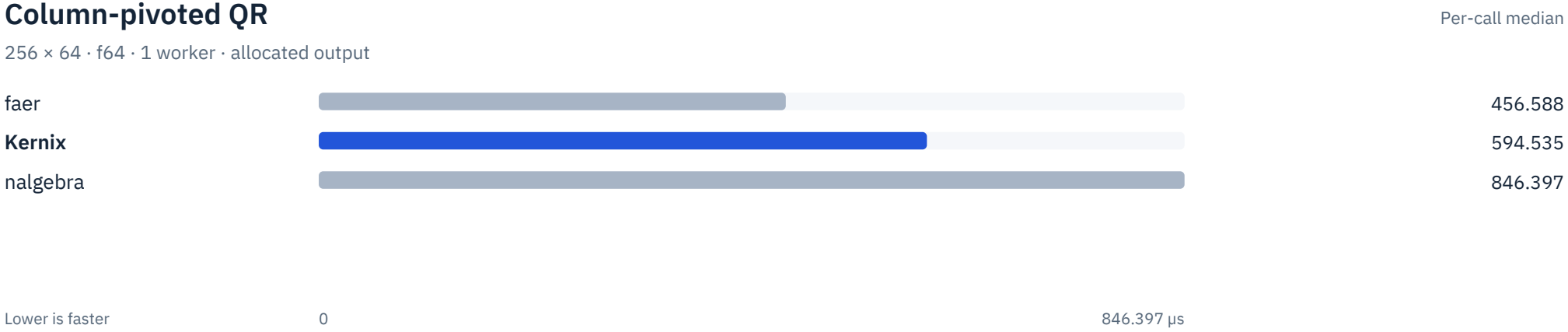
Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output



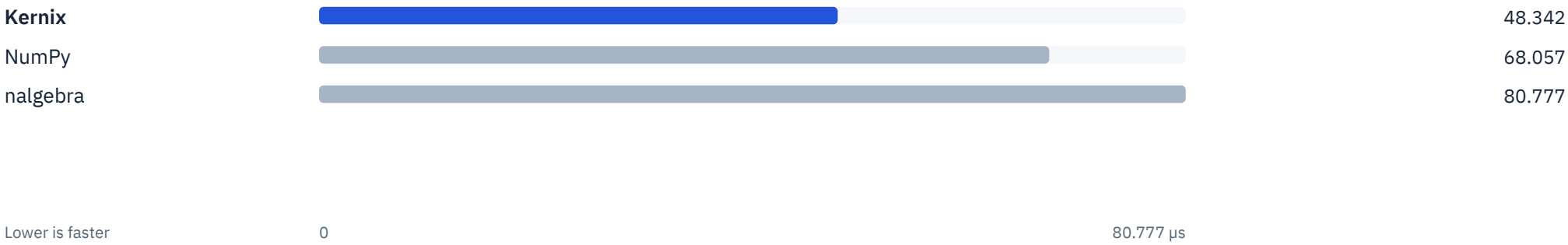
Kernix 0.13.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

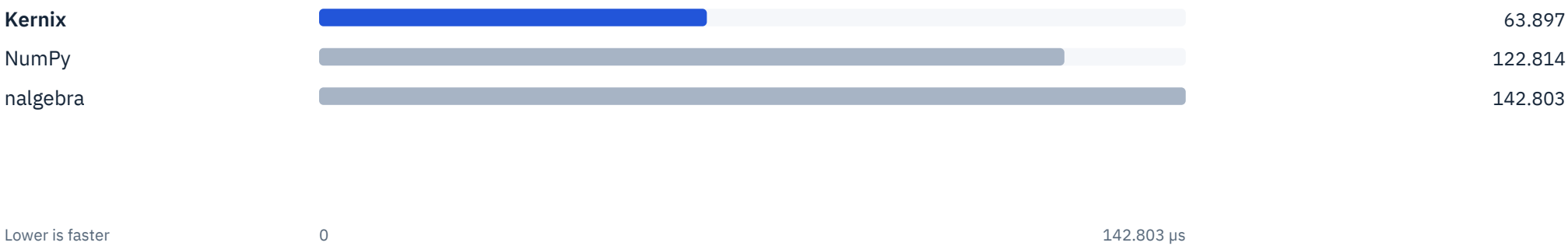
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

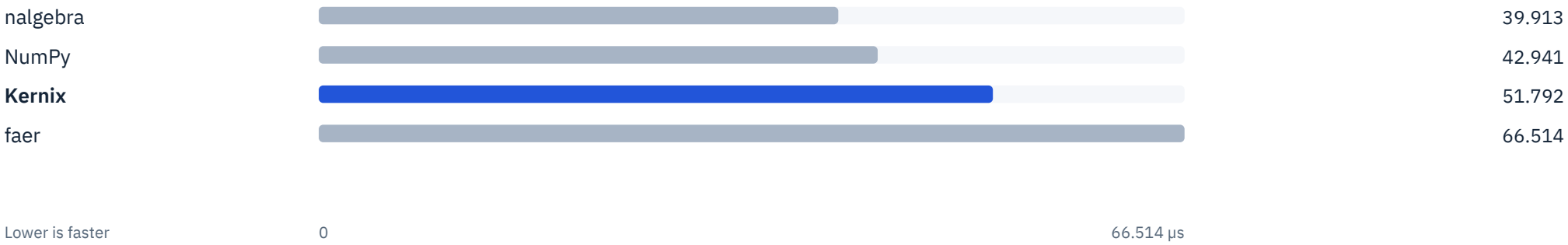


Kernix 0.13.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.13.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

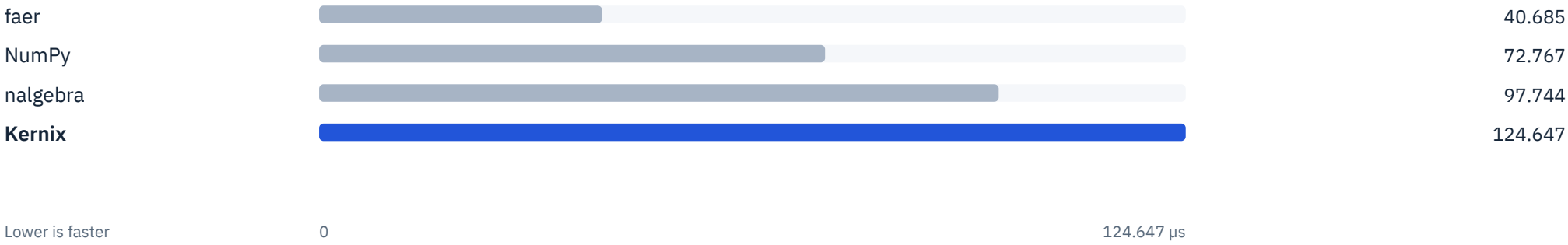
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Kernix 0.13.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.13.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.13.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

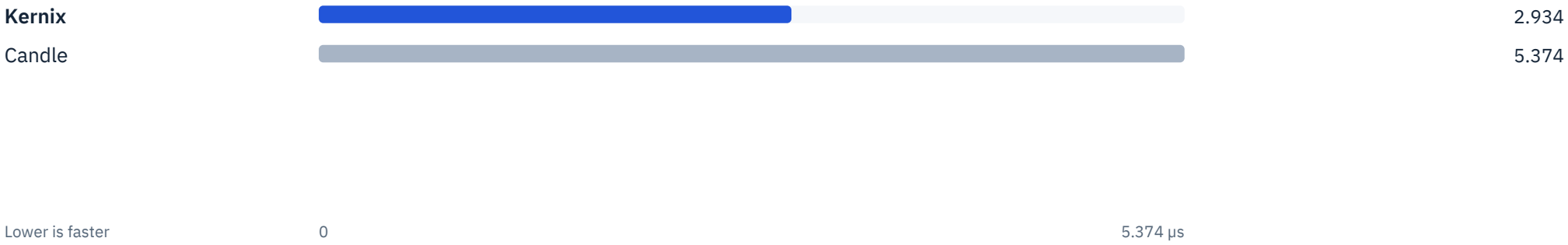
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.13.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



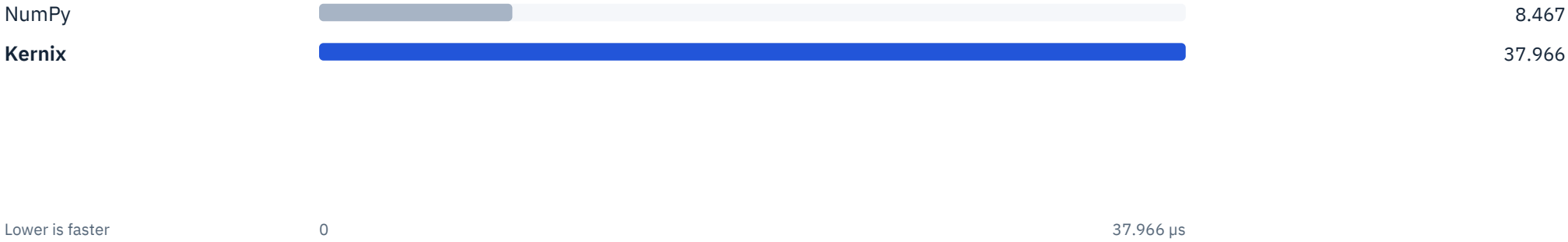
Kernix 0.13.0 / Tensor contractions

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.13.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



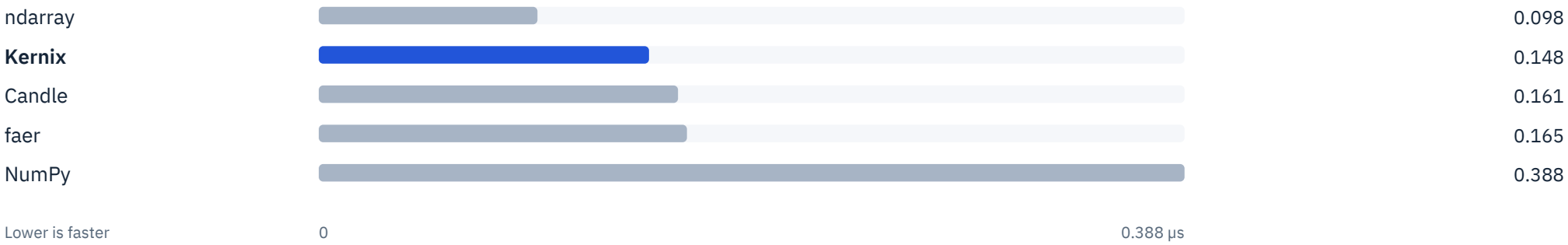
Kernix 0.13.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.971888310459063	0.9472371095915727 – 1.0012012012012013

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.998881793149922	0.9383370732247256 – 1.0158104873311955

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9928007590421402	0.8941054039275927 – 1.0236462883336268

Thin QR

512 × 128 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.015420789872515	1.0044781525261148 – 1.0341645738980965

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.994849890466581	0.9756770639941873 – 1.034197175088076

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9831306771759503	0.9551129339915351 – 1.003657321991281

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9934463407134071	0.9661845078310577 – 1.020309534524252

Thin QR

256 × 64 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9949400259294162	0.974955375897472 – 1.0243504436603224

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9724178403755869	0.9392735527809307 – 1.0354141656662665

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9959442169907882	0.9779128596239522 – 1.0724055236388552

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0037062955187517	0.9742674775091722 – 1.0601348384263685

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.011560810163837	0.9898909353651273 – 1.0408148018602503

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9931872157146883	0.9654477634052472 – 1.0037845234361267

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0118367346938775	0.9757423672103722 – 1.0385412349772067

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0067189532578082	0.9902451192773774 – 1.0252092008985458

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9839558786663324	0.952405309950067 – 1.032131599162936

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9970831307729704	0.975418410041841 – 1.1335920177383592

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0075329566854991	0.9835271317829457 – 1.0893697083725307

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9999751676185746	0.9566231767752397 – 1.0203783587570383

Cholesky factorization

256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0027173195152603	0.9950857348226192 – 1.0172574780149357

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9955066144074014	0.9943410264683163 – 1.0087164543796072

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9689889449139503	0.5479324905546913 – 1.0396271522059357

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.004840059681939	0.9558712930918805 – 1.0174677693830658

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9979826594557473	0.7234320374036477 – 1.0175336583620342

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse matrix × vector / faer 2048 × 2048 · five diagonals · f64	4938	4.938
Sparse matrix × vector / SciPy · CSR 2048 × 2048 · five diagonals · f64	7040	7.04
Sparse matrix × vector / NumPy 2048 × 2048 · five diagonals · f64	421739	421.739
Sparse matrix × vector / Kernix · explicit dense 2048 × 2048 · five diagonals · f64	3233102	3233.102
Sparse matrix × vector / Kernix 2048 × 2048 · five diagonals · f64	11708	11.708
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f64	22263	22.263
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f64	421951	421.951
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f64	428464	428.464
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f64	42111	42.111
Sparse × dense matrix / Kernix 512 × 512 · 32 RHS · five diagonals · f64	24034	24.034

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f32	15043	15.043
Sparse × dense matrix / Kernix 512 × 512 · 32 RHS · five diagonals · f32	16813	16.813
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f32	212719	212.719
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f32	169153	169.153
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f32	38575	38.575
Sparse × sparse matrix / NumPy 256 × 256 · five diagonals each · f64	635812	635.812
Sparse × sparse matrix / SciPy · CSR 256 × 256 · five diagonals each · f64	42096	42.096
Sparse × sparse matrix / Kernix · explicit dense 256 × 256 · five diagonals each · f64	573091	573.091
Sparse × sparse matrix / Kernix 256 × 256 · five diagonals each · f64	86747	86.747
Automatic CSR construction / Kernix 512 × 512 · clone + scan + compress · f64	534670	534.67

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Automatic CSR construction / SciPy · CSR 512 × 512 · clone + scan + compress · f64	941507	941.507
Automatic CSR construction / Kernix · explicit dense 512 × 512 · clone + scan + compress · f64	116698	116.698
Automatic CSR construction / NumPy 512 × 512 · clone + scan + compress · f64	81466	81.466
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	78262	78.262
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	114249	114.249
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	51166	51.166
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	40474	40.474
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	72444	72.444
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	76145	76.145
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	40065	40.065

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	37833	37.833
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1763	1.763
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1483	1.483
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2748	2.748
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1610	1.61
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1570	1.57
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1669	1.669
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	11606	11.606
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	11009	11.009
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	10383	10.383

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	11485	11.485
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	10478	10.478
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	12269	12.269
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	130215.00000000001	130.21500000000001
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	83555	83.555
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	87443	87.443
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	78164	78.164
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	118692	118.692
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	79583	79.583
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	619081	619.081

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	692011	692.011
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	640902	640.902
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	762531	762.531
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	576856	576.856
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	662704	662.704
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	6631	6.631
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	5167	5.167
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	5556	5.556
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	4990	4.99
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6203	6.203

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	5638	5.638
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	54080	54.08
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	53745	53.745
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	39874	39.874
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	38804	38.804
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	42827	42.827
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	39017	39.017
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	426304	426.304
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	341562	341.562
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	419397	419.397

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	303014	303.014
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	300744	300.744
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	284651	284.651
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2358	2.358
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3274	3.274
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1786	1.786
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	2575	2.575
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	1462	1.462
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1067	1.067
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	14817	14.817

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	41924	41.924
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	318499	318.499
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	41307	41.307
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	10516	10.516
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	4335	4.335
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	3939	3.939
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	227128	227.128
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1233339	1233.339
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	855367	855.367
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	1652097	1652.097

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	8215864	8215.864
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	8879196	8879.196
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	268121	268.121
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1157877	1157.877
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	882351	882.351
Cholesky rank-one update / Kernix 256 × 256 · f64	125067.99999999999	125.06799999999999
Cholesky rank-one update / nalgebra 256 × 256 · f64	24466	24.466
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	514277.99999999994	514.27799999999994
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	25679	25.679
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	560177	560.177

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky rank-one downdate / Kernix 256 × 256 · f64	116086	116.086
Cholesky rank-one update / nalgebra 512 × 512 · f64	152543	152.543
Cholesky rank-one update / Kernix 512 × 512 · f64	767096	767.096
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	3179033	3179.033
Thin QR / NumPy 512 × 128 · f64	2591262	2591.262
Thin QR / Kernix 512 × 128 · f64	2710334	2710.334
Thin QR / nalgebra 512 × 128 · f64	3063173	3063.173
Thin QR / faer 512 × 128 · f64	2074102	2074.102
Thin QR / nalgebra 1024 × 256 · f64	31419420.000000004	31419.420000000004
Thin QR / faer 1024 × 256 · f64	14910293	14910.293

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / Kernix 1024 × 256 · f64	18244794	18244.794
Thin QR / NumPy 1024 × 256 · f64	13106153	13106.153
Cholesky factorization / Kernix 512 × 512 · f64	3120465	3120.465
Cholesky factorization / NumPy 512 × 512 · f64	4526544	4526.544
Cholesky factorization / nalgebra 512 × 512 · f64	5562141	5562.141
Cholesky factorization / faer 512 × 512 · f64	1501998	1501.998
Column-pivoted QR / nalgebra 96 × 32 · f64	83846	83.846
Column-pivoted QR / Kernix 96 × 32 · f64	57355	57.355
Column-pivoted QR / faer 96 × 32 · f64	73177	73.177
Column-pivoted QR / Kernix 256 × 64 · f64	594535	594.535

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / nalgebra 256 × 64 · f64	846397	846.397
Column-pivoted QR / faer 256 × 64 · f64	456588	456.588
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	80777	80.777
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	48342	48.342
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	68057	68.057
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	63897	63.897
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	142803	142.803
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	122814	122.814
Thin QR / Kernix 96 × 32 · f64	51792	51.792
Thin QR / nalgebra 96 × 32 · f64	39913	39.913

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 96 × 32 · f64	42941	42.941
Thin QR / faer 96 × 32 · f64	66514	66.514
Thin QR / faer 256 × 64 · f64	424417	424.417
Thin QR / Kernix 256 × 64 · f64	510159.00000000006	510.15900000000006
Thin QR / NumPy 256 × 64 · f64	407643	407.643
Thin QR / nalgebra 256 × 64 · f64	393179	393.179
Cholesky factorization / faer 64 × 64 · f64	9145	9.145
Cholesky factorization / Kernix 64 × 64 · f64	20337	20.337
Cholesky factorization / nalgebra 64 × 64 · f64	14147	14.147
Cholesky factorization / NumPy 64 × 64 · f64	11160	11.16

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	124647	124.647
Cholesky factorization / faer 128 × 128 · f64	40685	40.685
Cholesky factorization / NumPy 128 × 128 · f64	72767	72.767
Cholesky factorization / nalgebra 128 × 128 · f64	97744	97.744
Cholesky factorization / nalgebra 256 × 256 · f64	721137	721.137
Cholesky factorization / faer 256 × 256 · f64	211038	211.038
Cholesky factorization / NumPy 256 × 256 · f64	538751	538.751
Cholesky factorization / Kernix 256 × 256 · f64	532935	532.935
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4211	4.211
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6659	6.659

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2397	2.397
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4177	4.177
Model inference / Candle batched normalized/sliced MLP · f64	3880	3.88
Model inference / Kernix batched normalized/sliced MLP · f64	2219	2.219
Model recording + forward / Candle batched normalized/sliced MLP · f64	5374	5.374
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2934	2.934
Model backward / Kernix prepared graph → fresh gradients · f64	3259	3.259
Model backward / Candle prepared graph → fresh gradients · f64	15270.000000000002	15.270000000000002
Parameter update / Kernix transactional SGD, fixed gradients · f64	553	0.553
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	8467	8.467

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	37966	37.966
Row softmax / ndarray 256 × 64 · f64	77091	77.091
Row softmax / NumPy 256 × 64 · f64	105888	105.888
Row softmax / Candle 256 × 64 · f64	87264	87.264
Row softmax / Kernix 256 × 64 · f64	93389	93.389
Create zeros / faer 64 × 64 · f32	165	0.165
Create zeros / Candle 64 × 64 · f32	161	0.161
Create zeros / ndarray 64 × 64 · f32	98	0.098
Create zeros / Kernix 64 × 64 · f32	148	0.148
Create zeros / NumPy 64 × 64 · f32	388	0.388

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create ones / faer 64 × 64 · f32	204	0.204
Create ones / ndarray 64 × 64 · f32	154	0.154
Create ones / Candle 64 × 64 · f32	249.00000000000003	0.24900000000000003
Create ones / NumPy 64 × 64 · f32	1330	1.33
Create ones / Kernix 64 × 64 · f32	191	0.191